Residual conductivity of a periodic Schrödinger operator with isotropic Dirac points

Abstract

We prove a residual conductivity formula for a periodic Schrödinger operator in two dimensions, describing non-interacting spinless electrons. Its Fermi surface consists of finitely many isolated, isotropic, twofold conical crossings. If there are M such crossings, the Cesàro mean of the current response to a constant electric field, with the linear-response limit taken first, is $M/16$ times the Euclidean conductivity tensor in atomic units. The proof derives the response from the Stark evolution, establishes an integrable band sum rule that cancels the term linear in time, and reduces each crossing to a two-dimensional matrix calculation. In particular, the assumptions on the band energies imply the required velocity matrix elements; these are not additional hypotheses.

1 Setting and theorem

Let $\mathcal{R} \subset \mathbb{R}^2$ be a lattice, let Ω be a fundamental cell, and let $\mathcal{B}$ be a fundamental cell of the reciprocal lattice

$$\mathcal{R}^* = \{K \in \mathbb{R}^2 : K \cdot R \in 2\pi\mathbb{Z} \text{ for all } R \in \mathcal{R}\}.$$

We identify quasimomenta modulo $\mathcal{R}^*$ and write $\mathbb{T}^2_* = \mathbb{R}^2/\mathcal{R}^*$. In particular,

$$|\Omega|\,|\mathcal{B}| = (2\pi)^2. \tag{1.1}$$

All spatial scalar products and norms are Euclidean. We use atomic units and consider one spin species.

For a real potential $V \in H^1_{\mathrm{per}}(\Omega)$, put

$$H = \tfrac{1}{2}p^2 + V, \qquad p = -\mathrm{i}\nabla, \qquad H_k = \tfrac{1}{2}(p+k)^2 + V. \tag{1.2}$$

Here H acts on $L^2(\mathbb{R}^2)$ and H_k acts on $\mathcal{H} = L^2_{\mathrm{per}}(\Omega)$, with domain $H^2_{\mathrm{per}}(\Omega)$. Its eigenvalues, counted with multiplicity, are $\lambda_1(k) \le \lambda_2(k) \le \cdots$; set $\lambda_0(k) = -\infty$.

Fix $\mu_{\mathrm{F}} \in \mathbb{R}$. The hypotheses are the following.

1. There are an integer $N \ge 1$ and distinct points $\mathcal{D} = \{k_1, \dots, k_M\} \subset \mathbb{T}^2_*$ such that

$$\lambda_N(k) < \mu_{\mathrm{F}} < \lambda_{N+1}(k) \qquad (k \in \mathbb{T}^2_* \setminus \mathcal{D}). \tag{1.3}$$

2. At each k_i, precisely two bands meet at the Fermi energy:

$$\lambda_{N-1}(k_i) < \lambda_N(k_i) = \mu_{\mathrm{F}} = \lambda_{N+1}(k_i) < \lambda_{N+2}(k_i). \tag{1.4}$$

3. For each i there is $v_i > 0$ such that, as $q \to 0$ in $\mathbb{R}^2$,

$$\begin{aligned} \lambda_N(k_i + q) &= \mu_{\mathrm{F}} - v_i|q| + O(|q|^2), \\ \lambda_{N+1}(k_i + q) &= \mu_{\mathrm{F}} + v_i|q| + O(|q|^2). \end{aligned} \tag{1.5}$$

The error bounds in (1.5) hold for all sufficiently small q, uniformly in its direction.

Let $e_\beta, e_\alpha \in \mathbb{R}^2$ denote the field and current directions; they need not be unit vectors. For $\varepsilon \geq 0$, define

$$\gamma_0 = \mathbf{1}_{(-\infty,\mu_{\mathrm{F}}]}(H), \qquad H_\varepsilon = H - \varepsilon x \cdot e_\beta, \qquad J_\alpha = p \cdot e_\alpha, \tag{1.6}$$

where H_ε is the self-adjoint closure of its restriction to $C_c^\infty(\mathbb{R}^2)$. This closure exists, as shown below. Set

$$\gamma_\varepsilon(t) = e^{-\mathrm{i}tH_\varepsilon}\gamma_0 e^{\mathrm{i}tH_\varepsilon}, \qquad j_\varepsilon(t) = \underline{\mathrm{Tr}}(J_\alpha \gamma_\varepsilon(t)), \qquad t \geq 0, \tag{1.7}$$

where, for the periodic locally trace-class operators in question,

$$\underline{\mathrm{Tr}}(A) = \frac{1}{|\Omega|}\,\mathrm{Tr}(\mathbf{1}_\Omega A \mathbf{1}_\Omega). \tag{1.8}$$

Theorem 1.1. *Under* (1.3)–(1.5)*, the current* (1.7) *is well defined for every* $\varepsilon, t \geq 0$*, and* $j_0(t) = 0$*. For every fixed* $t \geq 0$ *the limit*

$$\sigma_{\alpha\beta}(t) = \lim_{\varepsilon \downarrow 0} \frac{j_\varepsilon(t)}{\varepsilon}$$

exists. Its Cesàro mean satisfies

$$\lim_{T\to\infty} \frac{1}{T}\int_0^T \left(\lim_{\varepsilon\downarrow 0} \frac{j_\varepsilon(t)}{\varepsilon}\right) \mathrm{d}t = \frac{M}{16}\, e_\alpha \cdot e_\beta. \tag{1.9}$$

The theorem is conditional on the stated band structure. No lattice symmetry beyond periodicity is imposed, and no simplicity assumption is made on bands away from the Fermi crossings. The order of limits in (1.9) is part of the assertion.

2 Operator framework and the exact current

We first justify the dynamics, the trace, and the differentiations used later. Standard operator results are cited from Reed–Simon; the estimates needed at the conical crossings will be proved explicitly.

2.1 Domains and the Stark operator

Lemma 2.1. *The operator* H *is self-adjoint on* $H^2(\mathbb{R}^2)$*, and each* H_k *is self-adjoint on* $H^2_{\mathrm{per}}(\Omega)$ *with compact resolvent. For* k *in a compact set, its graph norm is uniformly equivalent to the* H^2 *norm. The family* $k \mapsto H_k$ *is analytic with this common domain. Moreover,* $H - \varepsilon x \cdot e_\beta$ *is essentially self-adjoint on* $C_c^\infty(\mathbb{R}^2)$ *for every real* ε*.*

Proof. Periodicity and $V \in L^2(\Omega)$ imply that V is uniformly locally L^2. The localization estimate of [2, Theorem XIII.96] gives, for every $\eta > 0$,

$$\|Vu\|_2 \leq \eta\|\Delta u\|_2 + C_\eta\|u\|_2. \tag{2.1}$$

The same estimate holds on the torus. Also,

$$|\langle u, Vu\rangle| \leq \eta\|\nabla u\|_2^2 + C_\eta\|u\|_2^2. \tag{2.2}$$

For completeness, on a bounded cell this follows from Hölder's inequality and the two-dimensional interpolation estimate $\|u\|_4^2 \leq C\|u\|_2(\|\nabla u\|_2 + \|u\|_2)$. A partition of unity by translates of bounded cells, with bounded overlap, proves the whole-space version. The Kato–Rellich theorem [1, Theorem X.12], applied also to $k \cdot p$, proves the domain and graph-norm

claims. Compactness follows from the compact embedding $H^2_{\mathrm{per}} \hookrightarrow L^2_{\mathrm{per}}$. The polynomial dependence on k, as a bounded map from H^2 to L^2, gives analyticity. Equivalently, a local Neumann series for the resolvent gives analyticity in the several variables k_1, k_2; see also [2, Section XII.2 and Section XIII.16].

Here are details for the unbounded electric potential. On $C_c^\infty(\mathbb{R}^2)$ introduce

$$\mathcal{N} = -\Delta + 2V + |x|^2 + b.$$

The harmonic oscillator $-\Delta + |x|^2 + 1$ has this space as a core by [1, Theorem X.28]. Its graph norm controls both Δ and $|x|^2$, as follows by integration by parts from

$$\|\Delta u\|_2^2 + \||x|^2 u\|_2^2 \le \|(-\Delta + |x|^2)u\|_2^2 + 4\|u\|_2^2.$$

Consequently (2.1) and Kato–Rellich show that $\mathcal{N}$ is self-adjoint on the oscillator domain and has the same core. Choose b large enough that $\mathcal{N} \ge 1$. By (2.2), its form controls $\|\nabla u\|_2^2 + \|xu\|_2^2 + \|u\|_2^2$. If $A = H - \varepsilon x \cdot e_\beta$, graph-norm equivalence gives

$$\|Au\|_2 \le C_\varepsilon \|\mathcal{N} u\|_2.$$

Since $\mathcal{N} = 2H + |x|^2 + b$, cancellation of the multiplication operators in the commutator gives, in the quadratic-form sense on the core,

$$[A, \mathcal{N}] = -\mathrm{i}(p \cdot x + x \cdot p) - 2\mathrm{i}\varepsilon p \cdot e_\beta.$$

It follows that

$$|\langle Au, \mathcal{N} u\rangle - \langle \mathcal{N} u, Au\rangle| \le C_\varepsilon \langle u, \mathcal{N} u\rangle.$$

The commutator theorem [1, Theorem X.37] now proves essential self-adjointness of A on C_c^∞. This argument does not differentiate the possibly unbounded potential V. □

We use the Bloch transform normalized by

$$(\mathcal{U} f)(k, x) = \sum_{R \in \mathcal{R}} e^{-\mathrm{i}k\cdot(x+R)} f(x+R), \qquad \mathcal{U} : L^2(\mathbb{R}^2) \longrightarrow \int_{\mathcal{B}}^{\oplus} \mathcal{H} \, \frac{\mathrm{d}k}{|\mathcal{B}|}. \tag{2.3}$$

It is unitary and decomposes H into H_k and J_α into

$$v_\alpha(k) = (p + k) \cdot e_\alpha = \partial_\alpha H_k, \qquad \partial_\alpha = e_\alpha \cdot \nabla_k. \tag{2.4}$$

These are the usual periodic-operator decompositions; see [2, Theorems XIII.97–XIII.99]. Under a reciprocal translation K, the fibers are conjugated by multiplication by $e^{-\mathrm{i}K\cdot x}$. All scalar traces below are therefore functions on $\mathbb{T}^2_*$.

Write

$$P(k) = \mathbf{1}_{(-\infty, \mu_{\mathrm{F}}]}(H_k). \tag{2.5}$$

It has rank N almost everywhere. At a point of $\mathcal{D}$ its actual rank is $N + 1$, but changing it on this finite set does not change any whole-space operator or integral. When convenient we choose there a commuting rank-N projection containing the lower bands. Uniform graph-norm estimates and the bounded occupied energies give

$$\sup_{k \in \mathcal{B}} \|P(k)\|_{L^2 \to H^2} < \infty, \qquad \sup_{k \in \mathcal{B}} \sum_{n=1}^{N} \|v_\alpha(k) u_n(k)\|_2^2 < \infty, \tag{2.6}$$

where $u_1(k), \dots, u_N(k)$ is any orthonormal occupied eigenbasis. No regularity of this choice of eigenvectors is asserted or needed.

2.2 Temporal gauge

Let $U_k^\varepsilon(t,s)$ be the propagator solving

$$\mathrm{i}\partial_t U_k^\varepsilon(t,s) = H_{k+\varepsilon t e_\beta} U_k^\varepsilon(t,s), \qquad U_k^\varepsilon(s,s) = 1. \tag{2.7}$$

The common-domain evolution theorem [1, Theorem X.70] applies: the domain is H^2_{per}, and the Hamiltonians and their time derivatives are continuous as maps $H^2_{\mathrm{per}} \to L^2_{\mathrm{per}}$, with uniformly equivalent graph norms on compact parameter sets. It gives a unitary propagator preserving H^2_{per}, with locally uniform bounds in that norm. The invertibility condition in that theorem is obtained by adding a constant so that the Hamiltonians are at least 1 on the parameter set in question, and then removing the resulting scalar phase. The identical argument on $\mathbb{R}^2$ gives the propagator $U^\varepsilon(t,s)$ for

$$H^{\mathrm{g}}_\varepsilon(t) = \tfrac{1}{2}(p + \varepsilon t e_\beta)^2 + V,$$

whose Bloch fibers are (2.7).

Lemma 2.2. *For $M_t = e^{\mathrm{i}\varepsilon t x\cdot e_\beta}$ one has*

$$e^{-\mathrm{i}tH_\varepsilon} = M_t U^\varepsilon(t,0). \tag{2.8}$$

The current is

$$j_\varepsilon(t) = \frac{1}{(2\pi)^2}\int_{\mathcal{B}} \mathrm{Tr}\big[(v_\alpha(k) + \varepsilon t(e_\alpha\cdot e_\beta))U_k^\varepsilon(t,0)P(k)U_k^\varepsilon(t,0)^*\big]\,\mathrm{d}k. \tag{2.9}$$

In particular, the localized operator defining this current is trace class for every finite ε, t.

Proof. The covariance $H^{\mathrm{g}}_\varepsilon(r+s) = M_s^* H^{\mathrm{g}}_\varepsilon(r) M_s$ and uniqueness of the propagator imply

$$U^\varepsilon(t+s,s) = M_s^* U^\varepsilon(t,0) M_s.$$

Thus $W(t) = M_t U^\varepsilon(t,0)$ is a strongly continuous unitary group. Its derivative at zero on C_c^∞ is $-\mathrm{i}(H - \varepsilon x\cdot e_\beta)$. By Lemma 2.1 its self-adjoint generator must be H_ε, proving (2.8).

Put $\rho^\varepsilon(t) = U^\varepsilon(t,0)\gamma_0 U^\varepsilon(t,0)^*$. This is periodic, with rank-N fiber projections

$$\rho_k^\varepsilon(t) = U_k^\varepsilon(t,0)P(k)U_k^\varepsilon(t,0)^*.$$

The operators $U_k^\varepsilon(t,0)P(k) : L^2 \to H^2$ are uniformly bounded on compact (ε,t) sets, by (2.6) and the propagator bounds. The density $\gamma_\varepsilon(t) = M_t\rho^\varepsilon(t)M_t^*$ is also periodic: conjugation of M_t by a lattice translation changes it only by a scalar phase.

We recall the trace normalization and explain the trace-class assertion. For a bounded periodic operator B with square-integrable Hilbert–Schmidt fibers, Bloch Parseval gives

$$\|\mathbf{1}_\Omega B\|^2_{\mathfrak{S}_2} = \frac{1}{|\mathcal{B}|}\int_{\mathcal{B}} \|B(k)\|^2_{\mathfrak{S}_2}\,\mathrm{d}k. \tag{2.10}$$

One can see this by inserting an orthonormal basis of functions supported in Ω into $B^*\mathbf{1}_\Omega$; its Bloch transforms are the same basis multiplied by $e^{-\mathrm{i}k\cdot x}$. Applying Parseval and then summing proves (2.10). The analogous identity holds for $B\mathbf{1}_\Omega$. The product of two such localized Hilbert–Schmidt operators is trace class, and polarization of the same identity gives the fiber trace formula for these products.

For a periodic projection Q of uniformly bounded fiber rank with $\sup_k \|Q(k)\|_{L^2\to H^1} < \infty$, both $\mathbf{1}_\Omega J_\alpha Q$ and $Q\mathbf{1}_\Omega$ are Hilbert–Schmidt. Consequently

$$\mathbf{1}_\Omega J_\alpha Q \mathbf{1}_\Omega = (\mathbf{1}_\Omega J_\alpha Q)(Q\mathbf{1}_\Omega)$$

is trace class. The same argument applies after the multiplication by M_t, which only shifts momentum by $\varepsilon t e_\beta$. For all operators used here it proves

$$\underline{\mathrm{Tr}}(A) = \frac{1}{|\Omega|\,|\mathcal{B}|}\int_{\mathcal{B}} \mathrm{Tr}\, A(k)\,\mathrm{d}k = \frac{1}{(2\pi)^2}\int_{\mathcal{B}} \mathrm{Tr}\, A(k)\,\mathrm{d}k. \tag{2.11}$$

Finally, M_t commutes with $\mathbf{1}_\Omega$ and $M_t^* J_\alpha M_t = J_\alpha + \varepsilon t(e_\alpha \cdot e_\beta)$. Unitary invariance of the localized trace, followed by (2.11), gives (2.9). □

3 A band sum rule at the Fermi energy

For $k \notin \mathcal{D}$, choose a measurable orthonormal eigenbasis $u_n(k)$ of H_k; such a choice is available as in [2, Theorem XIII.98]. For $n \le N < m$ define

$$\Delta_{mn}(k) = \lambda_m(k) - \lambda_n(k) > 0, \qquad c_{nm}^{\alpha\beta}(k) = \langle u_n, v_\alpha u_m\rangle\langle u_m, v_\beta u_n\rangle. \tag{3.1}$$

Inner products are conjugate linear in the first variable. Sums over complete eigenspaces are independent of the chosen basis, including at degeneracies within the occupied or unoccupied bands. Cauchy–Schwarz and (2.6) imply the useful uniform estimate

$$\sum_{n\le N<m} |c_{nm}^{\alpha\beta}(k)| \le \left(\sum_{n\le N} \|v_\alpha u_n\|_2^2\right)^{1/2}\left(\sum_{n\le N} \|v_\beta u_n\|_2^2\right)^{1/2} \le C_{\alpha\beta}. \tag{3.2}$$

In particular, the infinite sums below are absolutely convergent.

Let

$$F(k) = \sum_{n=1}^{N} \lambda_n(k), \qquad g(k) = \lambda_{N+1}(k) - \lambda_N(k). \tag{3.3}$$

Continuity of the bands, compactness of $\mathbb{T}^2_*$, and (1.5) give

$$g(k) \ge c\,\mathrm{dist}(k,\mathcal{D}) \quad \text{near } \mathcal{D}, \qquad \inf_{\mathrm{dist}(k,\mathcal{D})\ge\delta} g(k) > 0 \quad (\delta > 0). \tag{3.4}$$

Thus (3.2) implies

$$\int_{\mathcal{B}} \sum_{n\le N<m} \frac{|c_{nm}^{\alpha\beta}(k)|}{\Delta_{mn}(k)}\,\mathrm{d}k < \infty. \tag{3.5}$$

Proposition 3.1. *The function F is smooth off $\mathcal{D}$, has bounded first derivatives there, and satisfies*

$$\partial_\alpha F = \mathrm{Tr}(v_\alpha P), \tag{3.6}$$

$$\partial_\beta\partial_\alpha F = N(e_\alpha\cdot e_\beta) - 2\sum_{n\le N<m} \frac{\mathrm{Re}\, c_{nm}^{\alpha\beta}}{\Delta_{mn}}. \tag{3.7}$$

Both derivatives are integrable on the torus and have integral zero. In particular,

$$N|\mathcal{B}|(e_\alpha\cdot e_\beta) = 2\int_{\mathcal{B}} \sum_{n\le N<m} \frac{\mathrm{Re}\, c_{nm}^{\alpha\beta}}{\Delta_{mn}}\,\mathrm{d}k. \tag{3.8}$$

Also $j_0(t) = 0$.

Proof. Off $\mathcal{D}$, the occupied spectral cluster is isolated. The resolvent contour formula makes $P(k)$ analytic as a map $L^2 \to H^2$; this is the usual Riesz-projection argument of [2, Section XII.2]. Since $\partial_\beta P$ is off diagonal relative to $P \oplus (1-P)$, differentiating $F = \mathrm{Tr}(H_k P)$ proves (3.6). Its right-hand side is uniformly bounded by (2.6).

Differentiating $[H_k, P] = 0$ and $P^2 = P$ gives, for $n \le N < m$,

$$\langle u_m, (\partial_\beta P) u_n \rangle = -\frac{\langle u_m, v_\beta u_n \rangle}{\Delta_{mn}}.$$

The opposite off-diagonal block is its adjoint. Differentiating (3.6), using $\partial_\beta v_\alpha = (e_\alpha \cdot e_\beta)1$, proves (3.7). This proof uses the spectral projection and remains valid at internal band degeneracies.

The right-hand side of (3.7) is in L^1 by (3.5). Remove disjoint disks of radius δ around the points of $\mathcal{D}$ and apply the divergence theorem on the remaining torus. The boundary integral for $\partial_\beta \partial_\alpha F$ is bounded by $C\delta \sup |\partial_\alpha F|$, hence tends to zero. This proves that the integral of the Hessian entry is zero. The same argument for $\partial_\alpha F$, using boundedness of the continuous function F, proves that its integral is zero. In particular, the conical singularities create no missing point-supported term in the integrated sum rule.

At $\varepsilon = 0$, $U_k^0(t,0)$ commutes with $P(k)$, so (2.9) and (3.6) give $j_0(t) = (2\pi)^{-2} \int_{\mathcal{B}} \partial_\alpha F \, \mathrm{d}k = 0$. □

4 Linear response at fixed time

Proposition 4.1. *For every fixed $t \ge 0$ the linear response exists and is*

$$\sigma_{\alpha\beta}(t) = \frac{2}{(2\pi)^2} \int_{\mathcal{B}} \sum_{n \le N < m} \mathrm{Re}\, c_{nm}^{\alpha\beta}(k) \frac{\sin(t\Delta_{mn}(k))}{\Delta_{mn}(k)^2} \, \mathrm{d}k. \tag{4.1}$$

This formula is absolutely integrable for each finite t.

Proof. We first justify differentiation with the unbounded current observable. Fix a bounded time interval and take $|\varepsilon| \le 1$. For a unit vector u in the occupied subspace, set

$$\psi_\varepsilon(t) = U_k^\varepsilon(t,0)u, \qquad \psi_0(t) = e^{-\mathrm{i}tH_k}u.$$

Since

$$H_{k+\varepsilon s e_\beta} - H_k = \varepsilon s v_\beta(k) + \tfrac{1}{2}\varepsilon^2 s^2 |e_\beta|^2, \tag{4.2}$$

Duhamel's formula and the uniform H^2 bounds give

$$\|\psi_\varepsilon - \psi_0\|_2 \le C|\varepsilon|, \qquad \|\psi_\varepsilon\|_{H^2} + \|\psi_0\|_{H^2} \le C, \tag{4.3}$$

uniformly in k and u. Moreover,

$$\frac{\psi_\varepsilon(t) - \psi_0(t)}{\varepsilon} \longrightarrow \dot\psi(t) = -\mathrm{i} \int_0^t e^{-\mathrm{i}(t-s)H_k} s v_\beta(k) e^{-\mathrm{i}sH_k} u \, \mathrm{d}s \quad \text{in } L^2, \tag{4.4}$$

again uniformly for these parameters. To verify the uniformity without any regular choice of $u(k)$, Duhamel gives $\|U_k^\varepsilon(t,s) - U_k^0(t,s)\|_{H^2 \to L^2} \le C|\varepsilon|$. Interpolation with the L^2 bound gives an $O(|\varepsilon|^{1/2})$ bound from H^1 to L^2. The vector $v_\beta e^{-\mathrm{i}sH_k}u$ is uniformly bounded in H^1, which proves (4.4) by the Duhamel formula with $U_k^\varepsilon(t,s)$ on the left of the perturbation.

Writing $w = \psi_\varepsilon - \psi_0$, interpolation in (4.3) gives $\|w\|_{H^1} = O(|\varepsilon|^{1/2})$. The exact expansion

$$\langle \psi_\varepsilon, v_\alpha \psi_\varepsilon \rangle - \langle \psi_0, v_\alpha \psi_0 \rangle = 2\,\mathrm{Re}\langle w, v_\alpha \psi_0 \rangle + \langle w, v_\alpha w \rangle$$

has last term $O(|\varepsilon|^{3/2})$. It follows that the current expectations are differentiable at zero, uniformly in k. This also justifies differentiation under the integral in (2.9).

In density-matrix notation the derivative is

$$\dot\rho_k(t) = -\mathrm{i}\int_0^t s\, e^{-\mathrm{i}(t-s)H_k}[v_\beta(k), P(k)]e^{\mathrm{i}(t-s)H_k}\,\mathrm{d}s. \tag{4.5}$$

This expression may equally be read as the finite-rank sum obtained from (4.4). The free propagator preserves H^1, the form domain of H_k shifted by a constant, so the velocity traces in this formula are well defined.

For $\tau = t - s$, insertion of an eigenbasis in (4.5) gives

$$-\mathrm{i}\,\mathrm{Tr}\big(v_\alpha e^{-\mathrm{i}\tau H_k}[v_\beta, P]e^{\mathrm{i}\tau H_k}\big) = 2\sum_{n\le N<m} \mathrm{Im}\big(c_{nm}^{\alpha\beta} e^{-\mathrm{i}\tau\Delta_{mn}}\big).$$

Since

$$\int_0^t s\sin((t-s)\Delta)\,\mathrm{d}s = \frac{t}{\Delta} - \frac{\sin(t\Delta)}{\Delta^2}, \qquad \int_0^t s\cos((t-s)\Delta)\,\mathrm{d}s = \frac{1-\cos(t\Delta)}{\Delta^2},$$

the derivative of (2.9) is

$$\begin{aligned}\sigma_{\alpha\beta}(t) = \frac{1}{(2\pi)^2}\int_{\mathcal{B}}\Bigg\{ & tN(e_\alpha\cdot e_\beta) \\ & - 2\sum_{n\le N<m}\mathrm{Re}\, c_{nm}^{\alpha\beta}\left(\frac{t}{\Delta_{mn}} - \frac{\sin(t\Delta_{mn})}{\Delta_{mn}^2}\right) \\ & + 2\sum_{n\le N<m}\mathrm{Im}\, c_{nm}^{\alpha\beta}\frac{1-\cos(t\Delta_{mn})}{\Delta_{mn}^2}\Bigg\}\,\mathrm{d}k.\end{aligned} \tag{4.6}$$

All terms are integrable by (3.2), (3.5), and the bounds $|\sin(t\Delta)| \le t\Delta$ and $1-\cos(t\Delta) \le t^2\Delta^2/2$.

Realness of V now removes the last line of (4.6). Indeed, complex conjugation $\mathcal{C}$ satisfies

$$\mathcal{C}H_k\mathcal{C} = H_{-k}, \qquad \mathcal{C}v_\alpha(k)\mathcal{C} = -v_\alpha(-k). \tag{4.7}$$

Eigenvectors can be paired locally at k and $-k$ by $\mathcal{C}$. In these paired bases the gaps agree and $c_{nm}^{\alpha\beta}(-k) = \overline{c_{nm}^{\alpha\beta}(k)}$. For degenerate eigenvalues this statement is used after summing over the corresponding eigenspaces, so it is basis independent. The last line is therefore an odd integrable function on the torus and has integral zero.

Finally (3.8) cancels exactly the coefficient of t in (4.6). This proves (4.1). Its asserted integrability follows directly from $|\sin(t\Delta)|/\Delta^2 \le t/\Delta$ and (3.5). □

5 The matrix structure forced by an isotropic cone

Only one transition near each k_i can have a small gap. We now determine its leading velocity matrix element from the band hypotheses. Write $q = r\omega$, $r > 0$, $\omega \in S^1$, and set

$$\Delta_i(q) = \lambda_{N+1}(k_i+q) - \lambda_N(k_i+q), \qquad G_i^{\alpha\beta}(q) = \mathrm{Re}\, c_{N,N+1}^{\alpha\beta}(k_i+q). \tag{5.1}$$

Lemma 5.1. *Uniformly in ω as $r \downarrow 0$,*

$$\Delta_i(r\omega) = 2v_i r + O(r^2), \tag{5.2}$$

$$G_i^{\alpha\beta}(r\omega) = v_i^2\big(e_\alpha\cdot e_\beta - (e_\alpha\cdot\omega)(e_\beta\cdot\omega)\big) + O(r). \tag{5.3}$$

Proof. Fix a crossing and temporarily suppress i. By (1.4), a fixed resolvent contour around μ_{F} defines an analytic rank-two spectral projection $\Pi(q)$ for small q, including $q = 0$. It is analytic as a map $L^2 \to H^2$. Choose an isometry $W_0 : \mathbb{C}^2 \to \operatorname{Ran}\Pi(0)$ and define

$$W(q) = \Pi(q)W_0(W_0^*\Pi(q)W_0)^{-1/2}. \tag{5.4}$$

For small real q the matrix inside the square root is positive and close to the identity. Thus $W(q)$ is a real-analytic isometry into H^2, $W(q)W(q)^* = \Pi(q)$, and $W(0) = W_0$. This explicit frame is a local version of analytic spectral-subspace reduction; compare [2, Theorem XII.12]. It does not assert analyticity of the two individual bands at the crossing.

The Hermitian matrix

$$h(q) = W(q)^* H_{k_i+q} W(q)$$

has precisely the two crossing eigenvalues. With respect to fixed orthonormal coordinates in $\mathbb{R}^2$, Taylor expansion gives

$$h(q) = \mu_{\mathrm{F}} I + q_1 A_1 + q_2 A_2 + O(|q|^2), \qquad A_j = W_0^*(p_j + (k_i)_j)W_0. \tag{5.5}$$

To obtain the formula for A_j, differentiate h at zero and use $H_{k_i}W_0 = \mu_{\mathrm{F}}W_0$ and $\partial_j(W^*W) = 0$; the two frame-derivative terms cancel.

For any $\omega \in S^1$, the eigenvalues of $\omega_1 A_1 + \omega_2 A_2$ are $-v$ and v by (1.5). In particular, both A_j are traceless. In terms of the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

write $A_j = a_j \cdot \boldsymbol{\sigma}$, $a_j \in \mathbb{R}^3$. The eigenvalue condition says

$$|q_1 a_1 + q_2 a_2|^2 = v^2(q_1^2 + q_2^2), \qquad a_j \cdot a_\ell = v^2 \delta_{j\ell}.$$

A constant unitary change of basis in $\mathbb{C}^2$ therefore puts (5.5) into the form

$$h(q) = \mu_{\mathrm{F}} I + v(q_1\sigma_1 + q_2\sigma_2) + O(r^2). \tag{5.6}$$

One elementary way to see this change of basis is to diagonalize A_1/v, use $A_1A_2 + A_2A_1 = 0$ to make A_2/v off diagonal, and then adjust the phase of one basis vector. A further fixed unitary rotation gives the displayed pair of Pauli matrices.

Write $h(q) = a_0(q)I + d(q) \cdot \boldsymbol{\sigma}$. Equation (5.6) implies $d(q) = v(q_1, q_2, 0) + O(r^2)$, so the two eigenprojections obey

$$\pi_\pm(q) = \tfrac{1}{2}(I \pm \widehat{d}(q) \cdot \boldsymbol{\sigma}) = \tfrac{1}{2}(I \pm \omega_1\sigma_1 \pm \omega_2\sigma_2) + O(r). \tag{5.7}$$

All estimates are uniform in ω, because the Taylor remainders are uniform and $|d(q)| \geq cr$.

The compressed physical velocity, rather than the derivative of an arbitrarily chosen eigenvector, is the relevant observable. For any $e \in \mathbb{R}^2$ it satisfies

$$V_e(q) := W(q)^*((p + k_i + q) \cdot e)W(q) = v(e_1\sigma_1 + e_2\sigma_2) + O(r). \tag{5.8}$$

This follows directly from the H^2 analyticity of W and (5.5). Using (5.7) and (5.8),

$$G_i^{\alpha\beta}(q) = \operatorname{Re}\operatorname{Tr}_{\mathbb{C}^2}(\pi_-(q)V_{e_\alpha}(q)\pi_+(q)V_{e_\beta}(q)).$$

For $a, b, n \in \mathbb{R}^3$ with $|n| = 1$, Pauli multiplication gives

$$\operatorname{Re}\operatorname{Tr}\left[\frac{I - n \cdot \boldsymbol{\sigma}}{2}(a \cdot \boldsymbol{\sigma})\frac{I + n \cdot \boldsymbol{\sigma}}{2}(b \cdot \boldsymbol{\sigma})\right] = a \cdot b - (a \cdot n)(b \cdot n). \tag{5.9}$$

Taking $n = (\omega_1, \omega_2, 0)$ and $a = v(e_{\alpha,1}, e_{\alpha,2}, 0)$, $b = v(e_{\beta,1}, e_{\beta,2}, 0)$ proves (5.3). Formula (5.2) already follows from (1.5), or from $\Delta_i(q) = 2|d(q)|$. □

6 The long-time Cesàro limit

Fubini's theorem is applicable to (4.1) on every finite time interval, because its absolute value is dominated by t times the integrable sum in (3.5). Therefore

$$\frac{1}{T}\int_0^T \sigma_{\alpha\beta}(t)\,\mathrm{d}t = \frac{2}{(2\pi)^2}\int_{\mathcal{B}} \sum_{n\le N<m} \operatorname{Re} c_{nm}^{\alpha\beta}(k)\frac{1-\cos(T\Delta_{mn}(k))}{T\Delta_{mn}(k)^3}\,\mathrm{d}k. \tag{6.1}$$

Choose disjoint small disks $D_i = \{k_i+q : |q|<\delta\}$ in the quasimomentum torus. Shrink δ so that the two-band cluster is isolated on each disk and Lemma 5.1 applies. On the complement of these disks every occupied-to-unoccupied gap is bounded below by a positive constant. Within D_i, every such gap other than the one with $(n,m)=(N,N+1)$ is also bounded below by a positive constant: this is exactly the separation from λ_{N-1} and λ_{N+2} in (1.4). Compactness gives a common positive bound for all these remaining transitions. Thus (3.2) and $|1-\cos s|\le 2$ show that their entire contribution to (6.1) is $O(T^{-1})$.

It remains to evaluate, for each i,

$$I_i(T) = \frac{2}{(2\pi)^2}\int_{|q|<\delta} G_i^{\alpha\beta}(q)\frac{1-\cos(T\Delta_i(q))}{T\Delta_i(q)^3}\,\mathrm{d}q. \tag{6.2}$$

Change variables $q = z/T$. The Jacobian cancels the powers of T:

$$I_i(T) = \frac{2}{(2\pi)^2}\int_{\mathbb{R}^2} \mathbf{1}_{\{|z|<T\delta\}} G_i^{\alpha\beta}(z/T)\frac{1-\cos(T\Delta_i(z/T))}{(T\Delta_i(z/T))^3}\,\mathrm{d}z. \tag{6.3}$$

For each $z\neq 0$, Lemma 5.1 gives

$$T\Delta_i(z/T)\longrightarrow 2v_i|z|, \qquad G_i^{\alpha\beta}(z/T)\longrightarrow v_i^2\left(e_\alpha\cdot e_\beta-(e_\alpha\cdot\widehat{z})(e_\beta\cdot\widehat{z})\right).$$

Moreover, on the support of the integrand, $c|z|\le T\Delta_i(z/T)\le C|z|$ and $|G_i^{\alpha\beta}|\le C$. Since $1-\cos s\le \min(s^2/2,2)$, the absolute value of the integrand in (6.3) is bounded by

$$C\begin{cases}|z|^{-1}, & 0<|z|\le 1,\\ |z|^{-3}, & |z|>1.\end{cases} \tag{6.4}$$

This is integrable on $\mathbb{R}^2$. Dominated convergence is therefore applicable; in particular, it treats the nonlinear error in the gap inside the oscillatory factor without any differentiation of that factor.

Writing $z=r\omega(\theta)$ in the limiting integral gives

$$\begin{aligned}\lim_{T\to\infty} I_i(T) = \frac{2v_i^2}{(2\pi)^2}\int_0^{2\pi}&\left(e_\alpha\cdot e_\beta-(e_\alpha\cdot\omega(\theta))(e_\beta\cdot\omega(\theta))\right)\mathrm{d}\theta\\ &\times\int_0^\infty \frac{1-\cos(2v_ir)}{(2v_ir)^3}r\,\mathrm{d}r.\end{aligned} \tag{6.5}$$

The angular integral equals $\pi(e_\alpha\cdot e_\beta)$, because $\int_0^{2\pi}\omega_j\omega_\ell\,\mathrm{d}\theta=\pi\delta_{j\ell}$. The radial integral, with $s=2v_ir$, equals

$$\frac{1}{4v_i^2}\int_0^\infty\frac{1-\cos s}{s^2}\,\mathrm{d}s=\frac{\pi}{8v_i^2}.$$

For a direct verification of the last equality, Tonelli's theorem and $s^{-2}=\int_0^\infty ae^{-as}\,\mathrm{d}a$ yield

$$\int_0^\infty\frac{1-\cos s}{s^2}\,\mathrm{d}s=\int_0^\infty a\left(\frac{1}{a}-\frac{a}{a^2+1}\right)\mathrm{d}a=\int_0^\infty\frac{\mathrm{d}a}{1+a^2}=\frac{\pi}{2}.$$

Consequently all dependence on the cone speed cancels, and

$$\lim_{T\to\infty} I_i(T) = \frac{2v_i^2}{(2\pi)^2}\,\pi(e_\alpha \cdot e_\beta)\,\frac{\pi}{8v_i^2} = \frac{1}{16}\,e_\alpha \cdot e_\beta. \tag{6.6}$$

Summing (6.6) over the M crossings and adding the $O(T^{-1})$ remainder proves (1.9). Together with Lemmas 2.1 and 2.2, and Propositions 3.1 and 4.1, this completes the proof of Theorem 1.1. □

7 Consequences and scope

The value 1/16 is the contribution of a single cone for one spin species with the normalization (1.8) and (1.1). Thus two cones give 1/8 for the spinless model; two independent identical spin copies give 1/4. Each cone is counted once on the reciprocal torus, even if a drawing of a Brillouin cell places representatives on several boundary edges.

Time-reversal symmetry, supplied here by realness of V, is used to cancel the imaginary, antisymmetric part of the response in (4.6). Isotropy is used in (5.6)–(5.3); the cone velocities themselves need not be equal at different crossings.

The averaging in (1.9) produces the integrable kernel (6.4). The proof therefore requires no non-stationary phase assumption on the other bands and makes no claim that the unaveraged response converges at large time. Likewise, it does not interchange the small-field and long-time limits. If the Fermi gap is everywhere positive (the case of no crossings), the same proof gives zero Cesàro limit directly from (6.1).

References

[1] M. Reed and B. Simon, *Methods of Modern Mathematical Physics, II: Fourier Analysis, Self-Adjointness*, Academic Press, 1975.

[2] M. Reed and B. Simon, *Methods of Modern Mathematical Physics, IV: Analysis of Operators*, Academic Press, 1978.